	Academic Semester	Fall 2026-2027	**Academic level**	3
	Course Code	MT301	**Exam**	Quiz 1 — Model G
	Course Title	Probability and Statistics I		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Mai Farag , Dr. Walid Abd-Elraheem		

Student Name: _____ **Student ID:** _____

*Answer **all** questions. Calculators are allowed. Show your work in Questions 2 and 3; give probabilities as simplified fractions.*

Question 1 — Choose the correct answer (4 points)

One point each: ½ for the correct letter, ½ for a correct reason. Circle the letter and write why you chose it in the box.

1. A restaurant offers 6 starters, 3 main dishes and 2 desserts. How many meals with one choice from each category are possible?

A) 36 **B)** 11 **C)** 18 **D)** 38

Why? (reason or calculation)

2. A club has 12 members. In how many ways can it appoint a president, a vice-president and a secretary, with a different person in each post?

A) 220 **B)** 1728 **C)** 36 **D)** 1320

Why? (reason or calculation)

3. Two fair dice are rolled. The probability that the sum of their faces is 7 is

A) $\frac{1}{12}$ **B)** $\frac{1}{9}$ **C)** $\frac{1}{6}$ **D)** $\frac{1}{36}$

Why? (reason or calculation)

4. A fair coin is tossed four times. The probability of getting **at least one head** is

A) $\frac{3}{8}$ **B)** $\frac{15}{16}$ **C)** $\frac{1}{16}$ **D)** $\frac{1}{4}$

Why? (reason or calculation)

Question 2 — Counting (3 points)

A committee of 4 people is to be chosen from 8 men and 4 women.

(a) In how many ways can the committee be chosen? [1.5 pt]

(b) In how many ways can it be chosen if it must contain **exactly 1 women**? [1.5 pt]

Question 3 — Probability (3 points)

A 3-digit number is formed at random from the digits $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with **no digit repeated**.

(a) Find the probability that the number is **divisible by 5**. [1.5 pt]

(b) Find the probability that the number is **greater than 500**. [1.5 pt]

— End of quiz · Good luck —